

A PROTOCOL OF INTERMITTENT EXERCISE (SHUTTLE RUNS) TO TRAIN YOUNG BASKETBALL PLAYERS

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ABSTRACT

Zadro, I, Sepulcri, L, Lazzer, S, Fregolent, R, and Zamparo, P. A protocol of intermittent exercise (shuttle runs) to train young basketball players. *J Strength Cond Res* 25(6): 1767–1773, 2011—The purpose of this study was to set up a protocol of intermittent exercise to train young basketball players. Twenty-one players were asked to complete (a) an incremental test to determine maximal oxygen uptake ($\dot{V}O_{2\max}$), the speed at the ventilatory threshold (v_{thr}) and the energy cost of “linear” running (C_r) and (b) an intermittent test composed of 10 shuttle runs of 10-second duration and 30-seconds of recovery (total duration: about 6 minutes). The exercise intensity (the running speed, v) was set at 130% of v_{thr} . During the intermittent tests, oxygen uptake ($\dot{V}O_2$) and blood lactate concentration (La_b) were measured. The average pretraining $\dot{V}O_2$ calculated for a single bout ($131 \pm 9 \text{ ml} \cdot \text{min}^{-1} \cdot \text{kg}^{-1}$) was about 2.4 times greater than the subjects’ measured $\dot{V}O_{2\max}$ ($54.7 \pm 4.6 \text{ ml} \cdot \text{min}^{-1} \cdot \text{kg}^{-1}$). The net energy cost of running ($9.2 \pm 0.9 \text{ J} \cdot \text{m}^{-1} \cdot \text{kg}^{-1}$) was about 2.4 times higher than that measured at constant “linear” speed ($3.9 \pm 0.3 \text{ J} \cdot \text{m}^{-1} \cdot \text{kg}^{-1}$). The intermittent test was repeated after 7 weeks of training: 9 subjects (control group [CG]) maintained their traditional training schedule, whereas for 12 subjects (experimental group [EG]) part of the training was replaced by intermittent exercise (the same shuttle test as described above). After training, the $\dot{V}O_2$ measured during the intermittent test was significantly reduced ($p < 0.05$) in both groups (-10.9% in EG and -4.6% in CG %), whereas La_b decreased significantly only for EG (-31.5%). These data suggest that this training protocol is effective in reducing lactate accumulation in young basketball players.

KEY WORDS oxidative capacity, shuttle test, training intensity, running economy

INTRODUCTION

To reach top performance in basketball, players must be able to run different distances with continuous accelerations and decelerations, sudden stops and changes of direction, while performing correctly the specific actions of the game (dribbling, jumping, throwing-in, etc.) (13,18). Furthermore, they should be able to repeat these actions during the match and to use the short recovery periods between actions efficiently.

The ability to produce high levels of force and speed and to perform repeated high-intensity “bouts of action” during a match has become increasingly important in recent years because the level of amateur and professional basketball players has grown. Moreover, the number and duration of the “bouts of action,” respectively, increase and decrease with the technical level of the players: players under 19 years of age perform 55 ± 11 “bouts of action” during a match with an average duration of 2.1 seconds (2), whereas for professional players, the average number of “bouts of action” is 105 ± 52 , with an average duration of 1.7 seconds (18).

These data suggest (a) that increasing the capability of performing a higher number of high-intensity actions by means of more specific training is an important aim for this sporting activity; (b) that a basketball match could be considered as a type of high-intensity intermittent exercise; and therefore, (c) intermittent exercise could be considered a more specific tool to train basketball players, especially in the case of younger players with a lower technical level.

During intermittent exercise (and during a basketball match), the capability to recover between “bouts of action” depends on (a) the intensity and duration of the exercise bout and (b) the intensity and duration of the recovery period.

However, to plan a specific training protocol using intermittent training is rather complex because changing 1 or more of these factors (intensity and duration of exercise and recovery) elicits different metabolic responses (3,4,15).

As an example, it has been shown that the duration of the recovery period affects the relative use of the anaerobic and aerobic pathways (17). The minimal recovery time for which no lactate accumulation took place (after the first few runs) was between 20 and 30 seconds, a duration that corresponds approximately to half the time ($t/2 = 0.69\tau$) of the alactic

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oxygen deficit payment; with shorter pauses between bouts (<25 seconds) anaerobic glycolysis took place and blood lactate accumulated (17).

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Besides the duration of the recovery period, the intensity of exercise is an important factor in determining the metabolic response. Data reported by Gaitanos (12) indicate that if the energy expenditure is greater than that corresponding to the sum of $\dot{V}O_2$ during the bout and of the alactic fraction of the oxygen deficit, anaerobic glycolysis takes place and blood lactate accumulates even if the interval between bouts is sufficient to allow for a substantial phosphocreatine (PCr) resynthesis.

These data indicate that the interplay among intensity and duration of the bout and recovery phases establishes whether or not, after the first few bouts, lactate starts accumulating in the body fluids. It is also apparent that, to avoid lactate accumulation, intensity and duration of bout and recovery phases must be tailored to the subject's $\dot{V}O_{2\max}$ and muscle fiber type.

Indeed, it has been shown that $\dot{V}O_{2\max}$ is an important factor in determining the ability to carry out intermittent high-intensity exercise (5,6,21) and that the ability to buffer the H^+ ions produced by the anaerobic lactic metabolism is important to maintain the performance of repeated short bouts (6). A recent paper also indicates that lactate accumulation during a basketball game, varies greatly (from 3.7 to 13.2 mM) depending on the intensity of the game and on the role of the player (19).

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The aim of this study was, therefore, to set up a protocol of intermittent exercise to train young basketball players. In particular, we wanted to verify the hypothesis that with this training it would be possible to reduce the net accumulation of lactate in blood because this factor limits the capability of performing repeated "bouts of action," which are so typical of basketball matches. The exercise intensity, its duration, and the time of recovery of the training protocol were carefully planned to take into account the considerations reported above and to match as closely as possible the conditions during the game. Furthermore, the exercise intensity (in terms of running speed) was individually determined to take into account, as much as possible, intersubject differences.

METHODS

Experimental Approach to the Problem

In this study, we wanted to examine the effects of an intermittent training program on lactate accumulation and on energy expenditure. We asked young basketball players to complete an incremental test on a treadmill to determine maximal oxygen uptake ($\dot{V}O_{2\max}$) and the speed at the ventilatory threshold (v_{thr}). We then set up an intermittent test based on the individually determined values of v_{thr} : Each participant was asked to run a given distance (with a 180° change of direction) at a speed equal to 130% of v_{thr} and to repeat this exercise 10 times with 30-second recovery between bouts. Based on data reported in the

literature, this exercise intensity and this balance between exercise time and recovery time should be sufficient to induce training adaptations but not too high to induce a substantial lactate accumulation. After having demonstrated that this was indeed the case (the intensity of this type of exercise is more than twice that reached at $\dot{V}O_{2\max}$; however, lactate accumulation at the end of exercise is not maximal), we decided to use the same test as a training "tool." We divided our participants in 2 groups: 12 players in the experimental group (EG) for which part of the usual training was replaced by this intermittent exercise and 9 players in the control group (CG) who maintained their traditional training schedule. The training lasted 7 weeks after which we asked the subjects to repeat the intermittent test in the laboratory. Should the intermittent training be effective, the subjects of the EG should show better adaptations in terms of energy expenditure and of lactate accumulation in comparison to the CG.

Subjects

The experiments were performed on 21 young male basketball players whose anthropometric and physiological characteristics are given in Table 1. All subjects were recruited from the Snaidero Basket junior team (Italian basketball-league A) and had played basketball since they were 10 years old (5–7 years' experience).

Written informed consent was obtained from the participants and their parents. The study conformed to the standards set by the *Declaration of Helsinki*, and the

TABLE 1. Anthropometric characteristics of the subjects and physiological data obtained from the incremental test before training (at $W0$).^{*†}

	EG (n = 12)	CG (n = 9)
Age (y)	14.8 ± 0.6	17.6 ± 1.4‡
Body mass (kg)	77.7 ± 7.8	80.7 ± 11.0
Stature (m)	1.86 ± 0.08	1.91 ± 0.10
BMI (kg·m ⁻²)	22.4 ± 1.2	22.1 ± 2.1
$\dot{V}O_{2\max}$ (L·min ⁻¹)	4.18 ± 0.37	4.46 ± 0.51
HRmax (b·min ⁻¹)	191 ± 6	187 ± 8
C_r (J·m ⁻¹ ·kg ⁻¹)	3.98 ± 0.40	3.85 ± 0.21
$v_{\max}$ (m·s ⁻¹)	4.55 ± 0.20	4.60 ± 0.34
v_{thr} (m·s ⁻¹)	3.99 ± 0.32	4.23 ± 0.38
v_i (m·s ⁻¹)	5.19 ± 0.49	5.50 ± 0.49

^{*}EG = experimental group; CG = control group; BMI = body mass index; $\dot{V}O_{2\max}$ = maximal oxygen uptake; HRmax = maximal heart rate; C_r = net energy cost of running; $v_{\max}$ = speed at $\dot{V}O_{2\max}$; v_{thr} = speed at the ventilatory threshold; v_i = intermittent speed corresponding to 130% of v_{thr} .

[†]Values are mean ± SD.

[‡] $p < 0.05$ (1-way analysis of variance).

AU5 procedures were approved by the Ethics Committee (Institutional Review Board) of the local University.

Procedures

The participants were divided into 2 groups: 12 players in the EG and 9 players in the CG. Testing sessions were conducted just before the beginning (week 0, W0) and at the completion of a 7-week training program (week 7, W7). The testing sessions included assessment of anthropometric characteristics (body mass and stature) and physical capacities as indicated below.

Incremental Test. Maximal oxygen uptake ($\dot{V}O_{2\max}$) was determined at W0 by means of an incremental test on a treadmill (Saturn, HP Cosmos, Traunstein, Germany) with increments of $0.139 \text{ m}\cdot\text{s}^{-1}$ ($0.5 \text{ km}\cdot\text{h}^{-1}$) every 30 seconds (at an incline of 1°) until exhaustion (14). Before the beginning of the test, subjects were familiarized with the equipment and the procedures.

At the beginning of the test, 4 minutes of data was collected while the subjects ran at $2.78 \text{ m}\cdot\text{s}^{-1}$ ($10 \text{ km}\cdot\text{h}^{-1}$) to determine the oxygen consumption at submaximal speed ($\dot{V}O_{2\text{submax}}$).

During the experiments, heart rate (HR), oxygen consumption ($\dot{V}O_2$), CO_2 production ($\dot{V}CO_2$), minute ventilation ($\dot{V}E$), and respiratory exchange ratio (RER) were determined on a breath-by-breath basis by means of a previously calibrated metabolimeter (Quark b², Cosmed, Rome, Italy). The following parameters were thus obtained: (a) the $\dot{V}O_{2\max}$ and the speed at which $\dot{V}O_{2\max}$ was reached ($v_{\max}$); (b) the $\dot{V}O_2$, HR and speed at the ventilatory threshold ($\dot{V}O_{2\text{thr}}$, HR_{thr} , and v_{thr}); and (c) NET energy cost of running (C_r) from the ratio of net submaximal oxygen uptake to the speed (at $2.78 \text{ m}\cdot\text{s}^{-1}$): $C_r = (\dot{V}O_{2\text{submax}} - \dot{V}O_{2\text{bas}})/v$; C_r was expressed in $\text{J}\cdot\text{m}^{-1}\cdot\text{kg}^{-1}$ using an energy equivalent of $20.9 \text{ J}\cdot\text{mL}^{-1}$ of oxygen consumed.

At the end of the incremental test, a blood sample was obtained at the fourth and the sixth minute of recovery to determine blood lactate concentration (La_b) by means of a portable lactate analyzer (Lactate pro LT 1710, Arkray, Kyoto, Japan). The average value was used for further analysis.

Intermittent (Shuttle) Test. The participants were asked to run a given distance (see below), with a 180° change of direction at an individually determined speed. The intensity of exercise (the intermittent speed, v_i) was set equal to 130% of v_{thr} .

Each run lasted 10 seconds and was followed by 30 seconds of standing passive recovery. As an example, for $v_i = 4.44 \text{ m}\cdot\text{s}^{-1}$ (44.4 m in 10 seconds) the subject was asked to run 20 m, to make a turn of 180° and to cover 24.4 m in the opposite direction. The intermittent test lasted about 6 minutes (10 runs). During this test HR, $\dot{V}O_2$, $\dot{V}CO_2$, $\dot{V}E$, and RER were collected on a breath-by-breath basis by means of a previously calibrated portable metabolimeter (K4 b², Cosmed). The average values of HR, $\dot{V}O_2$, $\dot{V}CO_2$, $\dot{V}E$, and

RER recorded in the last 2 minutes of exercise (between the fourth and sixth minutes) were then calculated and used for further analysis.

At the end of the test, a blood sample was obtained at the fourth and the sixth minute of recovery to determine La_b by means of a portable lactate analyzer (Lactate pro LT 1710, Arkray). The average value was used for further analysis.

During these experiments, the effective speed attained during the test was monitored by means a photocell system (Racetime 2, Microgate System, Bolzano, Italy).

Training Protocol. Although the subjects were different in terms of age, their anthropometric and physiological characteristics did not differ (Table 1); moreover, all players belonged to the same team and were trained by the same coach. This allowed us to control (and match) their training volume and intensity. The 2 groups followed a similar 7-week training program (Tables 2 and 3).

The CG trained with 120 minutes of technical and tactical practice, 3 times per week without intermittent training. The EG group trained with 120 minutes of technical and tactical practice including 2–4 sets of 10 repetitions of intermittent running per training session, 3 times per week. A repetition consisted in 10 seconds of running (over a distance of about 40 m, see above) at v_i with a 180° change of direction (a shuttle run), separated by 30 seconds of “active recovery” (running at a speed of $2.22 \text{ m}\cdot\text{s}^{-1}$). Recovery between series was of 90 seconds (again at a speed of $2.22 \text{ m}\cdot\text{s}^{-1}$). The number of sets was progressively increased during the training period to maintain the training volume sufficiently high to induce training adaptations (Table 2). The training volume for each session and week was maintained the same for both groups (Table 3).

Statistical Analyses

Statistical analysis was performed using SPSS for Windows (SPSS Statistics 17.0) with significance set at $p \leq 0.05$. The effects of group, period (before, and after the 7 weeks of

TABLE 2. Training protocol for the experimental group.

Weeks of training	Days of training per week	Series	Repetitions
W1	3	2	10
W2	3	2	10
W3	3	2	10
W4	3	3	10
W5	3	3	10
W6	3	4	10
W7	3	4	10

training), and interaction (group * period) were tested using repeated-measures analysis of variance (ANOVA).

A 1-way ANOVA was used to assess differences between the 2 groups in the incremental test and in their anthropometric characteristics.

RESULTS

Anthropometric characteristics and data obtained from the incremental test before the 7-weeks' training program are shown in Table 1. Age was significantly lower in EG than in CG (1-way ANOVA, $p < 0.05$), but body mass, stature and BMI were not significantly different between groups. In addition, values obtained from the incremental test before training: $\dot{V}O_{2\max}$, HRmax, C_r , $v_{\max}$, v_{thrs} , and v_i were not significantly different between groups. The observed differences after the 7-week training period could therefore be attributed to the specificity of training and not to physical and physiological differences between the groups.

F1 Figure 1 shows a typical trace of oxygen uptake during an intermittent test: during and immediately after each bout of 10 seconds, $\dot{V}O_2$ increases sharply and decreases in the subsequent period of recovery. During this test, oxygen uptake reaches a "steady state" (i.e., the oscillations of oxygen uptake become more regular) after 1–2 minutes of exercise.

T4 The data of $\dot{V}O_2$ (net values, above resting conditions), $\dot{V}E$, HR, and RER collected between the fourth and sixth minutes of exercise during the intermittent test before training are reported in Table 4, together with the values of La_b , as measured at the end of the test, the measured values of v_i and the calculated values of C_r (see Discussion).

The statistical analysis (repeated-measures ANOVA) indicated no period effect (no differences between the pre and post measurements) and no differences between groups (and in the period $\times$ group interaction) for v_i , HR, RER, and $\dot{V}E$.

Significant pre to postdifferences were found for $\dot{V}O_2$ ($F(1,19) = 10.131$, $p < 0.05$) and C_r ($F(1,19) = 10.412$, $p < 0.05$) even if these parameters were found not to differ between groups (nor in the period * group interaction). These data indicate that both kinds of training (Table 3) result in improvements in submaximal $\dot{V}O_2$ and C_r in young basketball players but that intermittent training has no specific effect on these 2 parameters.

TABLE 3. An example of training practice in CG and EG: the distribution of 2 hours at W3.*

CG	EG
10-min warm-up	10-min warm-up
5-min Stretching	5-min Stretching
15-min Individual technical skill	15-min Individual technical skill
20-min Team technical skill	20-min Team technical skill
10-min Couple shooting	10-min Couple shooting
5-min Rest	5-min Rest
15-min 3 vs. 3 half court	15-min 3 vs. 3 half court
15-min 4 vs. 4 full court	15-min 4 vs. 4 full court
15-min 5 vs. 5 full court	15-min Intermittent exercise
10-min Cool down	10-min Cool down

*EG = experimental group; CG = control group; W3 = week 3.

The average values of La_b before and after training for the experimental and the CG are reported in Figure 2.

F2 This parameter was found to be significantly different between groups ($F(1,19) = 10.189$, $p < 0.05$); a significant difference was found for the period * group interaction ($F(1,19) = 5.174$, $p < 0.05$) but not for the pre to postcomparisons. A 1-way ANOVA applied to these data revealed that no

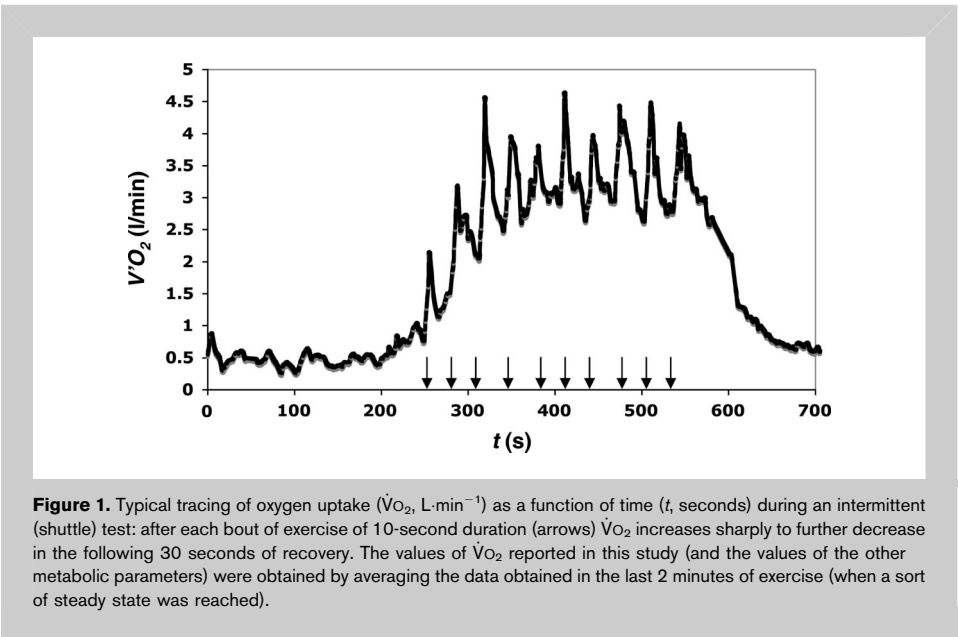


Figure 1. Typical tracing of oxygen uptake ($\dot{V}O_2$, L·min⁻¹) as a function of time (t , seconds) during an intermittent (shuttle) test: after each bout of exercise of 10-second duration (arrows) $\dot{V}O_2$ increases sharply to further decrease in the following 30 seconds of recovery. The values of $\dot{V}O_2$ reported in this study (and the values of the other metabolic parameters) were obtained by averaging the data obtained in the last 2 minutes of exercise (when a sort of steady state was reached).

TABLE 4. Data obtained from the intermittent (shuttle) test before (W0) and after (W7) the 7-week training program.*†‡

	EG (<i>n</i> = 12)		CG (<i>n</i> = 9)	
	W0	W7	W0	W7
v_i ($\text{m} \cdot \text{s}^{-1}$)	5.14 ± 0.32	5.17 ± 0.23	5.12 ± 0.45	5.10 ± 0.25
$\dot{V}\text{O}_2$ ($\text{ml} \cdot \text{min}^{-1} \cdot \text{kg}^{-1}$)	34.9 ± 4.0	31.1 ± 3.5	32.7 ± 2.9	31.2 ± 3.1
HR ($\text{b} \cdot \text{min}^{-1}$)	163 ± 4	164 ± 6	159 ± 8	165 ± 9
$\dot{V}\text{E}$ ($\text{L} \cdot \text{min}^{-1}$)	96.7 ± 14.3	90.7 ± 16.1	99.4 ± 19.0	98.7 ± 21.4
RER	1.12 ± 0.14	1.06 ± 0.10	1.12 ± 0.15	1.20 ± 0.12
La_b (mM)	7.34 ± 1.94	5.03 ± 1.98	8.27 ± 3.34	9.07 ± 2.13
C_r ($\text{J} \cdot \text{kg}^{-1} \cdot \text{m}^{-1}$)	9.45 ± 1.05	8.51 ± 1.28	8.92 ± 0.44	8.47 ± 0.47

*EG = experimental group; CG = control group; v_i = intermittent speed; $\dot{V}\text{O}_2$ = net oxygen uptake; HR = heart rate; $\dot{V}\text{E}$ = minute ventilation; RER = respiratory exchange ratio; La_b = lactate concentration in blood; C_r = net energy cost of running.

†Values are mean $\pm$ SD.

‡Values of $\dot{V}\text{O}_2$, HR, $\dot{V}\text{E}$, and RER are the average of the last 2 minutes of exercise. See text for details.

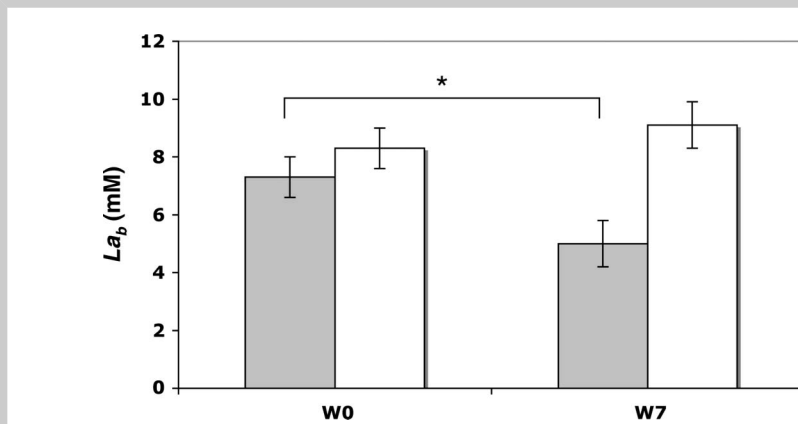


Figure 2. Average values of La_b (mM) before (W0) and after (W7) the 7-week training period for the Experimental (gray bars) and Control group (white bars). The asterisk indicates a significant ($p < 0.05$) decrease of La_b after training in the experimental group (of 3.7 ± 2.5 mM on the average).

significant differences between groups existed before training (at W0) but that the difference was highly significant posttraining (W7, $p < 0.001$). These data indicate that intermittent training (Table 3) is effective in reducing the amount of lactate accumulated after an intermittent test.

DISCUSSION

The most important outcome of this study is related to the observation that the values of La_b at the end of the intermittent test were indeed significantly reduced after specific training (in the EG). This finding, along with the observation that both $\dot{V}\text{O}_2$ and C_r decreased significantly after training, suggests that this specific protocol does indeed improve the oxidative capacity in basketball players.

Lactic acid accumulation in the body fluids is the outcome of an unbalance between lactate production and its oxidative removal (9). The oxidative removal of lactate, *ceteris paribus*, depends on the relative distribution of the oxidative and glycolytic fibers: The higher the percentage of type I fibers, the higher the oxidative capability of the subject and the faster the removal of lactate during the recovery phase of the intermittent exercise. The composition in fiber type plays an important role not only in the recovery phase but also at the onset of exercise when an O_2 deficit in contracted. This results from the

simultaneous contribution of 3 energy sources: oxygen stores, high-energy phosphates breakdown and early lactate accumulation (9). The higher the oxygen deficit, the lower the oxidative capacity: $\dot{V}\text{O}_2$ kinetics are faster (and the deficit lower) in subjects with a higher percentage of type I fibers in comparison to those with a higher percentage of type II fibers (1) and in trained subjects and subjects who are able to activate a higher percentage of slow twitch fibers at the beginning of the exercise (8). As a result, the higher the oxidative capability of the subject (the smaller the oxygen deficit) the lower the early lactate production and the faster the removal of lactate during the recovery phase of the intermittent exercise.

A conversion of intermediate fibers into the oxidative type after high-intensity intermittent training has been reported in the literature (7,16,20). Indeed, even if fast twitch fibers are

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recruited to a greater extent during intermittent exercise than during continuous exercise (11), the slow twitch fibers are more extensively involved during the rest periods for replenishing the PCr stores and for removing the lactate accumulated during the exercise periods.

These observations underline the importance of specific training for improving the oxidative capacity in basketball players who are required, during a match, to sustain repeated bouts of high-intensity exercise with frequent changes of direction and short recovery periods.

Another important observation is that, pretraining (at W0), the intersubject differences in La_b were quite large, ranging from 4 to 10 Mm. These differences could be attributed either to differences in lactate removal (mainly by oxidation to H_2O and CO_2 in skeletal muscle, see discussion above) or to differences in exercise intensity (and hence in lactate production, see discussion below). Indeed, besides large differences in La_b , we also found large intersubject differences in the energy requirement of this kind of exercise.

The latter can be calculated on the basis of (a) the net $\dot{V}O_2$ measured during the intermittent test ($\dot{V}O_2$, $ml \cdot min^{-1} \cdot kg^{-1}$) assuming that, in this condition, all the energy is derived from oxidative sources, either to sustain the energy needs during the bout or to replenish the PCr stores depleted during the exercise and (b) net lactate accumulation. (a) The energy derived from oxidative sources, per unit of time and per kilogram of body mass ($\dot{V}O_2$, $ml \cdot min^{-1} \cdot kg^{-1}$), is given by the following equation:

$$\dot{V}O_{2bout} = \dot{V}O_2 \cdot 40/10, \quad (1)$$

where 40/10 is the ratio between the overall duration of each "subunit" of an exercise cycle (a bout plus its recovery period: 40 seconds) and the duration of each exercise bout (10 seconds). Based on the average individual $\dot{V}O_2$ values (GC and EG) at W0, $\dot{V}O_{2bout}$ amounts to $131 \pm 9 ml \cdot min^{-1} \cdot kg^{-1}$, that is, about 2.5 times higher than the average $\dot{V}O_{2max}$ ($54.7 \pm 4.6 ml \cdot min^{-1} \cdot kg^{-1}$). The intersubject differences were quite high, ranging from 112 to 172 $ml \cdot min^{-1} \cdot kg^{-1}$.

This energy expenditure corresponds to an overall oxygen consumption of about $21.8 \pm 2.1 ml \cdot kg^{-1}$ during the 10 seconds bout ($\dot{V}O_{2bout}/60 \times 10$).

(b) The energy derived from anaerobic lactic sources can be calculated by dividing the net La_b accumulated at the end of the test (assuming La_b at rest equal to 1 mM) by the number of exercise bouts (10) and by assuming an energy equivalent of $3.3 ml \cdot kg^{-1} \cdot mM^{-1}$, as proposed by di Prampero and Ferretti (10). Based on data reported in Table 2, the anaerobic lactic contribution to the energy requirement amounts to $2.4 \pm 1.1 ml \cdot kg^{-1}$ (in O_2 equivalents), which brings the "overall" energy consumption (E) to $24.2 \pm 2.6 ml \cdot kg^{-1}$. These data indicate that at W0, about 10% of the overall energy requirement per bout is supplied by lactate, the remaining 90% being derived by oxidation and PCr breakdown (during the run) and resynthesis (during recovery). Thus, the average exercise intensity during the

10-seconds running bout (including La contribution) was about 2.7 times greater than the average $\dot{V}O_{2max}$ of the subjects.

The effective energy cost of running during the intermittent exercise (C_r) can be estimated from the average energy expenditure per bout (E) and from the average distance (d) covered during the 10-second bout (9):

$$C_r = E/d$$

Because the average v_i amounted to $5.14 \pm 0.3 m \cdot s^{-1}$ (weighted average of EC and CG), the average distance covered per bout was 51.4 m. Hence, the effective (net) energy cost of running during the intermittent test amounted to $24.2/51.4 = 0.471 ml \cdot kg^{-1} \cdot m^{-1}$ (or $9.2 \pm 0.9 J \cdot kg^{-1} \cdot m^{-1}$, using an energy equivalent of $20.9 J \cdot ml^{-1}$). This value is about 2.4-fold higher than the energy cost of running at constant "linear" speed on the flat, which, in our subjects, amounts to $3.9 \pm 0.3 J \cdot m^{-1} \cdot kg^{-1}$ (weighted average of EC and CG, see Table 1).

This difference depends on the fact that each 10 seconds of running comprised an acceleration phase (from zero to top speed) and a deceleration phase (from top speed to zero). Furthermore, every running bout was followed by a 180° turn. This scenario can indeed be expected to lead to an increase in the "overall" energy cost of running (C_r) and hence of the exercise intensity during the bout.

Thus, the large intersubject variability in $\dot{V}O_{2bout}$, La_b , and C_r can be attributed, among other things, to the capability of correctly and efficiently performing these phases of acceleration, deceleration, and to the "turning technique."

Thus, another reason why this specific training could have been useful in reducing the lactate accumulated at the end of the intermittent test is that the subjects learned to change direction more efficiently, thus leading to a reduction of the effective intensity of exercise. This hypothesis is supported by the observation that $\dot{V}O_2$ and $\dot{V}E$ (in EG) were lower at W7 with respect to W0 even if the speed of the shuttle runs (the "exercise intensity") was not different (Table 2). In any case, either because of an improved oxidative capacity or because of a better "biomechanical efficiency," the overall energy requirement was reduced in EG, indicating that this kind of training is indeed useful in improving performance in young basketball players.

Further studies on the effects of this kind of training on biomechanical factors (e.g., elastic recoil, efficiency of performance in the acceleration, and deceleration phases) could help in understanding the relative roles of biomechanical and physiological parameters.

In conclusion, data reported in this paper underline that, to reduce the energy demands during a basketball match, a player should be trained in such a way to minimize his energy cost of running, to optimize the biomechanics of the change of direction, and to maximize his oxidative capacity. Furthermore, our results suggest that the training protocol described here (with repeated bouts of high-intensity exercise, frequent changes of direction and short recovery

periods) could be useful in reducing lactate accumulation in basketball players.

PRACTICAL APPLICATIONS

When compared to regular training (5 vs. 5 practice), individual practice based on intermittent training has shown to be effective in reducing the lactate accumulated at the end of an intermittent test. This reduction should slow down the onset of peripheral fatigue and allow a player to perform a greater number of “action bouts” during a match. Data reported in this study also indicate that, to obtain these results, the intensity of the exercise should be tailored to the needs of each player (the speed of running should be proportional to the subject’s lactate threshold) and that the recovery time between the “bouts of action” should be carefully selected (30 vs. 10 seconds of exercise). Finally, the proposed training protocol included some of the specific gestures of this sport (accelerations and decelerations and changes of direction); this training is therefore expected to improve not only the economy but also the efficiency of movement, further improving performance in young basketball players.

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